

Fine-Tuning Cross-Encoders for Detecting Hallucinations in Financial Customer Service Chatbots

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Abstract

Mitigating hallucinations in Large Language Model (LLM) systems remains a key barrier to deployment in higher-risk domains such as finance and medicine where factual accuracy is both a regulatory requirement and component of responsible AI. This paper finetunes a cross-encoder DeBERTa-v3 transformer as a lightweight hallucination detection model to identify generated responses that are unsupported by retrieved evidence. The model is trained on Halubench benchmark data and a synthetic retail banking dataset. Evaluated against a LLM-as-a-judge, based on the Lynx framework Ravi et al. (2024), and a commercial AWS Bedrock contextual grounding guardrail, the model achieves up to 85% macro F1—outperforming Bedrock by 20–40 points on complex datasets and matching GPT-4o on structured tasks—while offering deterministic, low-latency, and transparent predictions. Results support its use as a scalable hallucination detection component in production systems.

1 Introduction

Large Language Models (LLMs) are increasingly deployed in customer-facing systems, supporting tasks from dialogue generation to decision assistance (Brown et al., 2020; OpenAI, 2023). Despite their fluency and versatility, LLMs frequently produce hallucinations—outputs that are unsupported by retrieved evidence (Ji et al., 2023). In high-risk domains such as finance, these errors can erode user trust, and introduce compliance and reputational risks.

Retrieval-Augmented Generation (RAG) is a commonly used technique for mitigating hallucinations by incorporating externally retrieved documents into the model prompt to support response grounding (Lewis et al., 2020). However, hallucinations persist even in RAG settings, often due to incomplete retrieval, poor integration of evidence, or continued reliance on a model’s intrinsic knowl-

edge. As a result, post-generation hallucination detection has become an important component of production-grade LLM systems, functioning as a safeguard or ‘guardrail’ before output delivery.

This paper aims to fine-tune a lightweight DeBERTa-v3-base cross-encoder as a guardrail for hallucination detection on the Halubench benchmark Ravi et al. (2024) and a synthetic financial-domain dataset simulating retail bank Q&A customer service interactions.¹ It compares this model against two baselines: an open-source LLM-evaluator based on the Lynx framework Ravi et al. (2024) and a close-sourced commercial contextual grounding guardrail provided by AWS Bedrock to assess its viability as a low-latency, scalable guardrail in production LLM-based systems for higher-risk domains like finance.

1.1 Background

Hallucination detection is typically framed as a *groundedness classification* task: for a given query, retrieved context, and model-generated response triplet, the guardrail must determine whether the response is supported by the evidence provided (Luo et al., 2024). Effective solutions must handle semantic nuance, domain-specific language, and variable response structures.

Several groundedness evaluation frameworks have emerged to address this challenge, with LLM-based approaches among the most widely adopted (Deng et al., 2023; Luo et al., 2024). RAGAS (Es et al., 2023), an open-source toolkit evaluates answer faithfulness by prompting an LLM to extract factual claims from a response and verify each claim against the retrieved context, returning a score based on the share of grounded claims.

Open-source LLM-as-a-judges, including both prompted and fine-tuned models, assess hallucinations by reasoning over the full query, context,

¹<https://huggingface.co/datasets/PatronusAI/HaluBench>

and response to provide a more holistic assessment of groundedness. Prompted approaches use general-purpose models such as GPT-4 to classify responses relative to retrieved context (Zheng et al., 2023; Manakul et al., 2023), while fine-tuned LLM-as-a-judges such as the open-source **Lynx** (Ravi et al., 2024) model are explicitly trained on hallucination datasets. Lynx has demonstrated stronger performance than GPT-4 on QA-style hallucination detection tasks.

Commercial LLM evaluators, such as **AWS Bedrock’s contextual grounding** guardrail, offer API-based hallucination checks for LLM outputs.

While effective, LLM-based evaluators present several practical limitations: they are costly to run at scale, introduce latency, and produce non-deterministic outputs—where repeated evaluations of the same input can yield different results. This limits reproducibility, complicates auditing, and creates inconsistencies in quality assurance workflows. These challenges motivate the need for efficient, locally deployable alternatives.

Encoder-based models offer a competitive alternative to LLM-based hallucination detection, combining transparency, low inference overhead, and real-time feasibility. **Luna** is a DeBERTa-large cross-encoder fine-tuned on multi-domain RAG data to detect hallucinated spans at the token level (Belyi et al., 2024). It outperforms GPT-3.5-based evaluators on the RAGTruth benchmark with a 97% cost reduction and 91% lower latency. **Lettuce Detect**, built on the ModernBERT architecture, surpasses Luna and GPT-4-turbo with a 79.2% F1—despite being 30× smaller than leading LLMs. (Ádám Kovács and Recski, 2025).

1.2 Contributions

This paper builds on this recent work aimed at addressing key limitations of LLM-based models. The contributions of this paper are:

1. Develop a practical, resource-efficient cross-encoder approach for hallucination detection in RAG systems that balances performance and latency;
2. Construct a domain-specific evaluation set reflecting the language and groundedness challenges of retail bank dialogue systems, leveraging the **Banking77 dataset** to enrich task realism and intent coverage²;
3. Conduct a comparative evaluation of the cross-encoder against both open-source and commercial LLM-based evaluators, highlighting trade-offs in groundedness accuracy, transparency, and deployability.

Together, these findings offer actionable guidance for operationalising hallucination detection in higher-risk environments such as banking, where LLM outputs must be both reliable and auditable.

2 Methodology

2.1 Data Construction

The training and evaluation data were sourced from two key datasets: (i) the HaluBench benchmark dataset; and (ii) a synthetic dataset of customer service queries for a retail-banking chatbot.

HaluBench Benchmark

This benchmark dataset contains around 15,000 query–context–response triplets annotated for hallucination detection. Introduced by Ravi et al. (2024) to develop the LYNX model,

HaluBench combines datasets spanning multiple domains and includes both naturally occurring and synthetic hallucinations.³ The benchmark consists of the following domain-specific subsets:

- **FinanceBench**—financial reasoning (Islam et al., 2023)
- **DROP**—numerical reasoning (Dua et al., 2019)
- **PubMedQA**—biomedical question answering (Jin et al., 2019)
- **RAGTruth**—retrieval-augmented generation evaluation (Wu et al., 2023)

Two sub-datasets—COVID-QA (Möller et al., 2020) and HaluEval (Li et al., 2023) datasets—were excluded from analysis due to lengthy report-style contexts and limited domain relevance, and due to a systematic length bias between grounded and hallucinated responses, respectively.

A final set of 3,900 observations were sampled from Halubench for both fine-tuning and evaluation (with 3,315 for training and 585 for the hold-out evaluation set).⁴

banking77

²<https://huggingface.co/datasets/PolyAI/>

³See Section 3.2 of Ravi et al. (2024) for further detail on

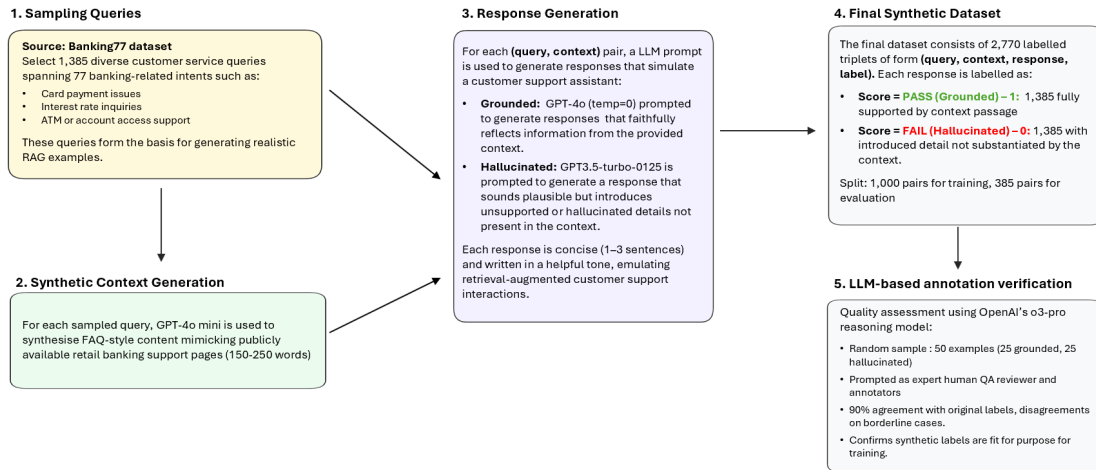


Figure 1: Pipeline for synthetic dataset construction of RetailBankQA

RetailBankQA Synthetic Dataset

To fine-tune the model in the context of retail bank customer service, a synthetic RetailBankQA dataset was constructed using the multi-stage prompting pipeline in Figure 1.

- **Step 1: Query Sampling.** A representative sample of 1,385 customer queries were obtained from the [Banking77 dataset](#), covering 77 intents (e.g., card_payment_problems, atm_support, interest_rate_query).⁵
- **Step 2: Context Generation.** For each query from Step 1, gpt-4o-mini was prompted to generate a 150–250 word FAQ-style passage that could plausibly be used to answer a customer’s banking query. Synthetic FAQ-style contexts were generated using LLMs rather than scraped banking help pages, due to commercial, ethical, and reproducibility concerns with using real institutional content.
- **Step 3: Response Generation.** Two responses created for each query-context pair:
 - **Grounded responses** generated using gpt-4o (temperature = 0), prompted to

the construction of the HaluBench benchmark dataset.

⁴The reduction in sample size compared to the full HALUBENCH dataset reflects the exclusion of the HALUEVAL subset (approximately 10,000 examples), which was removed due to a systematic length bias between hallucinated and grounded responses.

⁵The Banking77 dataset contains 13,083 real-world customer service banking queries and is widely used for research in intent classification and conversational AI.

rely strictly on the context passage without using external knowledge.

- **Hallucinated responses** were generated using gpt-3.5-turbo-0125, with prompting designed to subtly introduce factual inconsistencies (e.g., fabricated service features or incorrect fees) while maintaining plausible relevancy.

- **Step 4: Dataset Assembly.** The final dataset included 2,770 labeled (query, context, response) triplets—1,385 grounded and 1,385 hallucinated. Of these, 1,000 grounded–hallucinated pairs were allocated for training, and 385 pairs were held out as an evaluation set. The train–validation split was performed using unique pair identifiers to ensure no query–context–response overlap or data leakage across splits. An example test case is provided in Appendix A.
- **Step 5: LLM-based Annotation Verification.** To assess label quality, a random sample of 50 synthetic examples (25 grounded, 25 hallucinated) were reviewed by OpenAI’s o3-pro model⁶, prompted as an expert QA annotator. Given its stronger reasoning capabilities than the model used during data construction, o3-pro’s 95% agreement with the original labels suggests that the synthetic annotations are broadly consistent with expert judgment and suitable for training.

⁶<https://platform.openai.com/docs/models/o3>

Final sample

The final sample consists of 6,600 observations, with 3,900 from HaluBench and 2,770 test cases from the constructed RetailBankQA dataset.

Domain	# Examples	%H	Med CT	Max CT
RetailBankQA	2,770	50%	275	352
HaluBench	3,900	34%	291	2494
DROP	1,000	50%	224	1204
FinanceBench	1,000	50%	454	2494
PubMedQA	1,000	50%	280	937
RAGTruth	900	18%	282.5	538

Table 1: Summary statistics of full training and evaluation dataset, where %H denotes share of hallucinated responses and CT indicates context token length.

Table 1 reports token-level statistics across all datasets. The share of hallucinations are balanced in all datasets but RAGTruth. Each dataset was split using stratified sampling into training, validation, and test subsets (70/15/15) to preserve class balance. All examples were tokenized using the DeBERTa-v3-base tokenizer.

3 Model

The model used for binary hallucination classification is a fine-tuned cross-encoder based on the **DeBERTa-v3-base** architecture.⁷ This transformer has demonstrated performance on natural language inference (NLI) benchmarks such as MNLI, ANLI, and SNLI, outperforming models like BERT, RoBERTa, and XLNet (He et al., 2021; Liu et al., 2019; Yang et al., 2019).

DeBERTa-v3-base is well-suited for hallucination detection tasks that require fine-grained factual consistency checks, due to its pretraining on entailment-style objectives—that is, tasks where the model learns to assess whether one sentence is logically supported by another. Compared to bi-encoders or prompt-based LLM evaluators, the cross-encoder architecture jointly encodes the *query*, *retrieved context*, and *response*, enabling the model to capture rich inter-segment dependencies.

3.1 Training Setup

To adapt the pretrained DeBERTa-v3-base model, a classification head is added to produce a single logit representing the probability that the response is grounded. Inputs are tokenized in the format:

[CLS] query [SEP] context [SEP] response [SEP]

⁷<https://huggingface.co/microsoft/deberta-v3-base>

Training is performed using the Hugging Face Trainer API based on three epochs, a batch size of 8, a learning rate of 1×10^{-5} , and the AdamW optimizer (weight decay = 0.01). This configuration was selected based on a targeted grid search, where it achieved the most stable validation performance and highest F1 score, while reducing overfitting and training variance associated with smaller batch sizes.⁸ Binary cross-entropy loss is applied, with labels encoded as 1 for grounded (PASS) and 0 for hallucinated (FAIL). The training dataset is balanced and composed of examples from both HaluBench (general domain) and RetailBankQA (financial domain). The Hugging Face DeBERTa tokenizer is used to tokenize all inputs.

3.2 Context Window Constraints

To address the 512-token limit of DeBERTa-v3-base, we apply a windowed chunking strategy: inputs are split into overlapping chunks, evaluated independently, and aggregated via max-pooling to yield an example-level groundedness score. Token length analysis (Figure 2) shows that 94% of test cases fall within the limit. While token-level span attribution was considered, it was not implemented due to the added complexity and its limited benefit outside FinanceBench (where 44.3% of examples exceed 512 tokens, compared to less than 2.5% in other subsets). This approach follows Luna (Belyi et al., 2024) in chunking long RAG inputs but opts for coarse-grained classification to reduce inference cost. As model context windows expand, finer-grained span attribution remains a direction for future work.

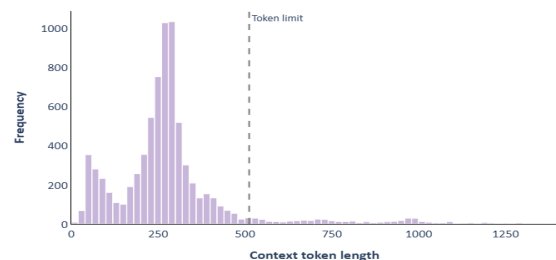


Figure 2: Context Token Length Distribution for full dataset (dashed line = 512-token limit).

4 Baseline Comparison Models

Baseline comparisons include a prompted GPT-4o evaluator and a commercial grounding guardrail

⁸Hyperparameter search results available at GitHub.

from AWS Bedrock.

4.1 Prompted LLM Verifier

A zero-shot evaluation baseline is implemented using OpenAI’s gpt-4o-2024-05-13 model, accessed via the OpenAI API.⁹ The prompting approach is adapted from the Lynx framework proposed by Ravi et al. (2024) and is presented using a structured Jinja2 template. The full prompt is included in Appendix B. Each input is formatted as a (query, document, response) triplet, and the model is instructed to assess whether the response is grounded in the retrieved document, based on strict faithfulness criteria.

The model returns a JSON object containing a REASONING field (a brief justification) and a SCORE field, which is parsed as either "PASS" or "FAIL". These binary outputs are used for groundedness classification in the evaluation.

4.2 AWS Bedrock Guardrail

As an off-the-shelf comparator model, the commercial contextual grounding guardrail from Amazon Bedrock was selected for its alignment with the contextual grounding task and its accessibility as a managed service.¹⁰ Each (query, context, response) triplet is submitted via the apply_guardrail API with contextual grounding policies enabled and relevance checks disabled. The system uses Claude Sonnet 4 (Anthropic, 2024), a foundation model available via AWS at the time of evaluation, to return a grounding score and a threshold-based decision.

A fixed threshold of 0.5 is applied to determine whether the response is grounded. Outputs are mapped to binary PASS or FAIL labels to support direct comparison with other evaluators.

4.3 Evaluation

Model evaluation was conducted on two test sets: the HaluBench hold-out set and the synthetic RetailBankQA test set. The classifier produces chunk-level predictions, which are then aggregated to the example level using the maximum predicted probability across all chunks for each test case.

Performance is reported using example-level accuracy, precision, recall, and F1-score, with PASS indicating a grounded response and FAIL indicating a hallucinated one. Results are further stratified by

source domain (e.g., FinanceBench, DROP) to assess generalization under domain shift. Qualitative failure cases are reviewed to understand where the model struggles with subtle hallucination patterns or ambiguous references.

5 Results and discussion

Results in Figure 3 show a comparative evaluation of the cross-encoder classifier and two LLM-based evaluators. Metrics include macro F1 scores, macro precision, macro recall, and accuracy.

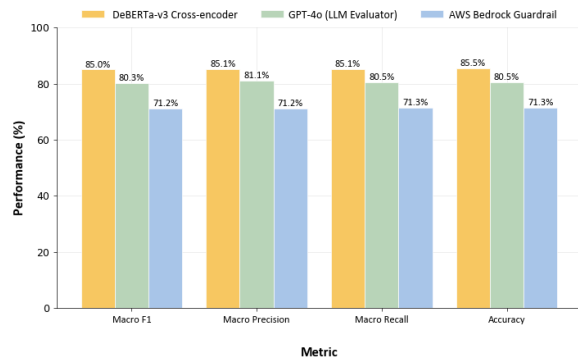


Figure 3: Performance comparison of cross-encoder versus LLM-based models; by evaluation metric.

The cross-encoder performs consistently across all metrics, with a macro F1 score of around 85%, compared to 80% for the GPT-4o evaluator and 71% for the AWS Bedrock guardrail. This result is influenced by the cross encoder’s performance on the RetailBankQA dataset (which may be relatively easy for this model, discussed below). However, these results still shows that a lightweight model can offer competitive predictive power relative to LLM-based alternatives. A more detailed breakdown of results show that the GPT-4o (LLM evaluator) demonstrates the most stable performance across the HaluBench sub-datasets and the RetailBankQA synthetic set, with macro F1 scores ranging from 76% and 85%. By contrast, the AWS Bedrock guardrail consistently underperforms with lower overall scores, and elevated false negative rates, particularly on DROP and PubMedQA. Its proprietary nature limits transparency, making it difficult to conduct any further analysis on underlying drivers.

The results reveal a clear capability gap between LLM-based and encoder-based methods. The GPT-4o evaluator demonstrates semantic abstraction, numeric reasoning, and broad domain coverage. It can accurately label paraphrased or free-

⁹<https://platform.openai.com/docs/models/gpt-4o>

¹⁰<https://docs.aws.amazon.com/bedrock/latest/userguide/guardrails-contextual-grounding-check.html>

Dataset	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
HaluBench	DeBERTa-v3 Cross-encoder	69.9%	69.8%	67.6%	67.8%
	GPT-4o (LLM Evaluator)	82.5%	82.9%	83.5%	82.5%
	AWS Bedrock	58.6%	60.5%	60.3%	58.5%
- DROP	DeBERTa-v3 Cross-encoder	88.0%	88.0%	88.1%	88.0%
	GPT-4o (LLM Evaluator)	82.4%	82.7%	82.2%	82.2%
	AWS Bedrock	47.9%	46.7%	47.1%	45.8%
- FinanceBench	DeBERTa-v3 Cross-encoder	52.2%	52.5%	52.0%	49.3%
	GPT-4o (LLM Evaluator)	84.8%	86.1%	84.9%	84.7%
	AWS Bedrock	50.6%	51.1%	50.8%	47.3%
- RAGTruth	DeBERTa-v3 Cross-encoder	75.8%	60.0%	60.0%	60.0%
	GPT-4o (LLM Evaluator)	84.7%	74.8%	77.2%	75.8%
	AWS Bedrock	80.6%	69.9%	74.7%	71.6%
- pubmedQA	DeBERTa-v3 Cross-encoder	68.6%	69.4%	67.2%	67.1%
	GPT-4o (LLM Evaluator)	77.9%	82.0%	79.4%	77.6%
	AWS Bedrock	60.0%	62.2%	61.3%	59.6%
RetailBankQA	DeBERTa-v3 Cross-encoder	96.6%	96.7%	96.6%	96.6%
	GPT-4o (LLM Evaluator)	79.0%	85.2%	79.0%	78.0%
	AWS Bedrock	80.9%	85.5%	80.9%	80.3%

Table 2: Evaluation results across six datasets comparing cross-encoder, LLM-based, and AWS Bedrock hallucination detection methods. **Note:** Accuracy is sensitive to class imbalance, and some datasets are not balanced between hallucinated and non-hallucinated cases. Macro-averaged metrics (precision, recall, F1) provide a more robust measure of detection performance in these settings.

Type	Description	% of FNs
Numeric reasoning & extraction	Cross-encoder struggles when interpreting numbers, ratios and financial calculations needed to assess groundedness. It also sometimes has difficulty with number formatting difficulty (e.g. rounding)	52%
Narrative summaries/paraphrase	Cross-encoder misses on open-ended, paraphrased, or summary-style responses, often due to low lexical overlap	26%
Verdict-based outcome responses	Cross-encoder labels vary from ground truths where completely different wording has been used to deliver an outcome-based response using evidence in context (e.g. biomedical verdicts)	22%

Table 3: Summary of dominant cross-encoder false negative types (mutually exclusive assignment).

form responses without lexical overlap, as seen in RAGTruth and PubMedQA, and shows strong numerical reasoning in FinanceBench and DROP, including normalization and basic calculations. However, GPT-4o can be overly permissive on datasets like RetailBankQA, often passing topically relevant summaries that lack grounding in the retrieved context. In contrast, the cross-encoder flags such vague responses as hallucinations due to its finer semantic alignment—highlighting the need for LLM evaluator calibration. The DeBERTa-v3 cross-encoder performs well on structured, template-driven tasks with short factual answers (e.g., RetailBankQA, DROP), where deterministic scoring aligns with consistent patterns. Its performance drops on examples requiring abstraction or multi-step reasoning. A manual audit of cross-encoder false negatives (Table 3) shows that 52% stem from numeric rea-

soning or value extraction failures, especially in financial calculations. Another 26% involve missed paraphrases or summary-style answers with low lexical overlap, and 22% involve verdict-style outputs (e.g., biomedical claims) where correct but differently worded answers are not matched.

6 Conclusion

Overall, these results show that the open-source LLM-evaluator provides a more flexible mechanism for hallucination detection on diverse, semantically complex, and numerically intensive dialogue settings. By contrast, cross-encoder models remain competitive in highly structured, factoid-heavy domains but struggle with abstraction, paraphrasing, and calculations. These findings support a hybrid approach, with cross-encoders serving as efficient first-stage filters for predictable queries and LLM-based evaluators handling ambiguous or novel cases.

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A **RetailBankQA Triplet with Grounded and Hallucinated Responses**

QUERY <i>How Long Will My Transaction Take?</i>
CONTEXT When making a transfer, the time it takes to reach the recipient can vary depending on the type of transfer. Faster Payments usually arrive within minutes, while BACS payments can take up to three working days. International transfers may take longer depending on the destination and receiving bank.
RESPONSE (Grounded) The timing for your transaction to be completed depends on the transfer type. Faster Payments typically arrive within minutes, BACS transfers can take up to three working days, and international transfers may take longer depending on the recipient's bank.
RESPONSE (Hallucinated) When you send a transfer, it will always arrive instantly, regardless of the method or country. All transactions are processed in real time by default, including international ones.
ANNOTATION Span: "it will always arrive instantly" and "including international ones" Reason: Original: "Faster Payments usually arrive within minutes... BACS... up to three working days... international transfers may take longer" Generative: "all transactions are processed in real time... including international ones"

Table 4: Example (query, context, response) triplet from RetailBankQA, showing a grounded and hallucinated response

B **Groundedness Evaluation Prompt**

The following prompt was used for LLM-based groundedness evaluation (GPT-4o). Query, context, and response fields were dynamically filled for each example.

Given the following QUESTION, DOCUMENT and ANSWER you must analyze the provided answer and determine whether it is faithful to the contents of the DOCUMENT.

A faithful ANSWER must not offer new information beyond the context provided in the DOCUMENT.

A faithful ANSWER must be grounded in the DOCUMENT and not based on external knowledge.

A faithful ANSWER also must not contradict information provided in the DOCUMENT.

Even if there is a slight discrepancy between the ANSWER and the DOCUMENT, or any ungrounded assumptions made in the ANSWER, the ANSWER must be considered not faithful.

The faithfulness of the ANSWER is independent of its relevancy to QUESTION.

Even if the ANSWER is irrelevant to the QUESTION, your task is to evaluate if it is faithful to the DOCUMENT regardless of relevancy.

Output your final verdict by strictly following this format: "PASS" if the answer is faithful to the DOCUMENT and "FAIL" if the answer is not faithful to the DOCUMENT. Briefly, show your reasoning.

QUESTION (THIS DOES NOT COUNT AS BACKGROUND INFORMATION):

```
{{user_query}}
-
DOCUMENT:
{{retrieved_context}}
-
ANSWER:
{{model_answer}}
-
Your output should be in JSON format with the
keys "REASONING" and "SCORE":
{"REASONING": <brief reasoning as bullet points>,
"SCORE": <your final score>}
```